

Agent Skills

Agentic Coding for Geospatial

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What is a Skill?

- Skills are a set of instructions that teach an AI agent how to perform specific, multi-step workflows.
- Skills are defined by a SKILL.md containing natural-language instructions for the agent along with specific examples and patterns.
- You can also publish your skill on GitHub and make it available for others to download and use with their own agent.

Skills are just Markdown files

- A folder containing SKILL.md file with instructions on how to complete task.
- The folder may also contain scripts and templates referred by the SKILL.md file.
- Skills are automatically loaded when a prompt or user intent matches the description of the skill.
- Optionally, you can invoke the skill using a command - using the skill name. i.e. /create-cog

```
create-cog > SKILL.md > ...
1  ---
2  name: create-cog
3  description: Convert a raster to a Cloud-Optimized GeoTIFF (COG)
4  ---
5
6  Use this skill when you need to convert any geospatial raster data to a
7  Cloud-Optimized GeoTIFF.
8
9  ## Workflow
10
11  1. Activate the conda environment that has GDAL. Never use the base environment.
12  2. Check the GDAL version, since the conversion command differs by version.
13  3. Check if the input is already a valid COG. If it is, stop and report
14  | that. Do not re-convert.
15  4. Convert, using the compression settings and naming convention below.
16  5. Validate the output and report the result.
17
18  ## Required Tools
19
20  * `gdal` Conda package. Install using `conda install -c conda-forge gdal`
21  * `rio-cogeo` Python package for validation. Install using `pip install
22  | rio-cogeo`
23
24  Both must come from a conda environment, not base. Check what is available
25  before installing:
26
27  | conda env list
28  | conda activate <env>
29  | gdal --version
30
31  If neither is installed, ask the user which environment to install into
32  before proceeding.
```

When to use Skills?

- Skills range for simple instructions to complex workflows.
- Use skills to change or guide model behavior.
 - Many users have created skills with instructions for the AI model to tailor its output for specific purposes. [[Example](#)]
- Use skills to automate a workflow.
 - Take any repeated multi-step task and create a skill to automate it. [[Example](#)]
- Use skills to ensure adherence to a specific coding pattern.
 - Encapsulate your preferences with examples [[Example](#)]

Using a Skill

Installing Skills

- Use Claude Code Plugins
 - Skills can be installed as plugins
- Install Skills manually
 - Copy the SKILL.md file to the appropriate folder
 - Ask Claude Code to copy the source file and install it as a Skill.
- Use npx skills manager
 - [npx skills](#) is an open standard for installing and managing skills.
 - Supports many proprietary and open agents
 - Requires Node.js

Creating a Skill

How to write good Skills?

- Clearly mention when the skill should be invoked.
 - You may also use negative triggers so the agent knows when not to use a skill
- Keep the skill body small
 - You can put deeper references in secondary files and link them from the main skill file.
 - i.e. Use the appropriate template from the template/ directory when you need one
- Outline a clear completion criteria
 - Add validation rules to ensure the task is completed correctly.
- Focus on edge cases and specific constraints
 - Add your domain knowledge to guide the agent on how to avoid common mistakes

Useful Skills

- [Anthropic Official Skills](#): Official skills repository by Anthropic. skill-creator, pdf, xlsx are useful skills for Data Science work.
- [Skills Leaderboard](#): Discover, install and manage skills using the skills CLI.
- [Geospatial Skills for Agents](#): A catalog of installable geospatial tools and capabilities designed for coding agents, covering raster/vector processing, satellite data, and interactive visualization.
- [SpatialThoughts AI Skills](#): Repository containing skills accompanied with Spatial Thoughts Courses.
- [GeoSQL](#): Skill for data scientists and analysts working with geospatial data on PostGIS, BigQuery, Snowflake, and Wherobots.